

Computer Networks

CS - (Fall 25)

Assignment-1

Marks 40

Submission *deadline* **12th October 2025 (11:59 PM)**

Instructions:

- *This is an individual Assignment. Demos will start soon after submission date, **missing demo will result in straight zero marks.***
 - ***All parties involved in any kind of plagiarism/cheating (even in a single line of code) will be given zero marks in all the assignments.***
 - ***Assignment deadline won't be extended***
 - ***Late submissions will be discarded so submit your assignment on-time***
 - *You must follow below submission guidelines, otherwise your submission won't be accepted*
 - *create a new directory*
 - *change its name to the following format*
YOURSECTION_ROLLNUMBER_NAME.
E.g. A_20I-0012_Muhammad Ahamd
 - *put all your files into this newly created directory (code files and output screenshots, scan pdf for theoretical questions)*
 - *Compress the directory into a compressed .zip file*
 - *Submit it on google classroom*
 - *Use good programming practices (well commented and indented code; meaningful variable names, readable code etc.).*
 - *Understanding the problem is also part of the assignment.*
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QUESTION 1 (10 Marks)

You are part of a team at Fast Tech Solutions, a tech startup implementing a smart office management system to improve workplace efficiency. The system uses IoT devices and sensors, allowing employees to interact with office services through a mobile app. Your task is to develop this system using UDP socket programming in C++, with both a server and multiple client programs.

Your task is to create the following components:

1. A server-side application that listens for multiple client requests and manages office services.
2. Client applications representing employees interacting with the system.

The server should:

- Accept connections from up to 5 simultaneous clients.
- Receive each employee's details (name, arrival time, location, temperature preference, and service request).
- Process the request and respond accordingly.
 - Example: If an employee requests a meeting room and sets the temperature to 26°C, the server should reply:
"details noted: meeting room, temp 26 from Sana on 2nd floor"
- Maintain a log of attendance and preferences for all connected clients (you may use an array, file writing, or any other logging mechanism).

The client should:

- Connect to the server using UDP sockets.
- Collect the user's input (name, arrival time, location, preferred temperature, and requested service).
- Send the details to the server.
- Receive the server's confirmation message and display it to the user.

System Requirements:

- The server must be capable of handling concurrent requests from 5 clients.
- Attendance and preferences of each client must be recorded and stored for later access by HR.

QUESTION 2 (10 Marks)

You are part of a team developing an online gaming platform and have been tasked with creating a simple multiplayer game called "Rock, Paper, Scissors" using socket programming in C++. The implementation should include both a server and client program. Your task is to create the following components:

1. A server-side application that listens for client connections and manages game state.
2. Two client applications, each representing one of the players.

The server should:

- Accept two simultaneous connections from clients.
- Prompt both connected clients to simultaneously send their choices (rock, paper, or scissors).
- Determine the winner based on standard game rules.
- Send the outcome to both clients and close connections.

The client should:

- Connect to the server.
- Prompt the user for their choice and send it to the server.
- Receive results from the game and display them.

Game Rules

Rock, Paper, Scissors rules:

- Rock beats scissors.
- Scissors beat paper.
- Paper beats rock.

QUESTION 3 (5 Marks)

1. Capture Live HTTP, FTP, and DNS Packets (at least 5 packets)

- Use **Wireshark** to filter and capture packets for:
 - HTTP GET requests
 - DNS Query responses
 - FTP file transfer data

2. Analyze a captured HTTP request

- Identify the requested URL, User-Agent, and cookies.
- Explain how HTTPS protects sensitive data compared to HTTP.

QUESTION 4 (10 Marks)

Consider that you are working on one of the lab systems of GoodWell software house having a LAN network with 350 Mb/s transmission rate to institutional Router 1, for which there is an institutional network connected to the Internet. Suppose that the average object size is 250,000 bits and that the

average request rate from the Lab's browsers to the origin servers is 35 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side (Router 2) of the access link forwards an HTTP request until it receives the response is 2.5 seconds on average. Model the total average response time as the sum of the average access delay (that is, the delay from Router 2 to Router 1) and the average Internet delay.

i. Find the total average response time. [4]

ii. Now suppose that a web cache is installed on the GoodWell Lab network's LAN. Suppose that the miss rate is 0.7. Find the total response time (considering that the response time from the cache is 0.01 sec) [6]

QUESTION 5 (5 Marks)

Consider a router with a buffer of 50,000 packets. Each packet = 1,000 bytes. Outgoing link = 100 Mbps. During a sudden burst, packets arrive at 150 Mbps for 10 seconds.

Tasks:

1. Compute how long it will take before the buffer overflows.
 2. Estimate how many packets will be dropped.
 3. Suggest two advanced mechanisms (beyond "increase buffer size") that real networks use to avoid such losses.
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